

COSC264 · INTRODUCTION TO COMPUTER NETWORKS AND THE INTERNET · TERM 4

Module 2

Routing and Addressing at Internet Scale

1 · **Inter-AS routing and AS policy**

2 · Intra-AS routing protocols

3 · Aggregation, NAT and IPv6

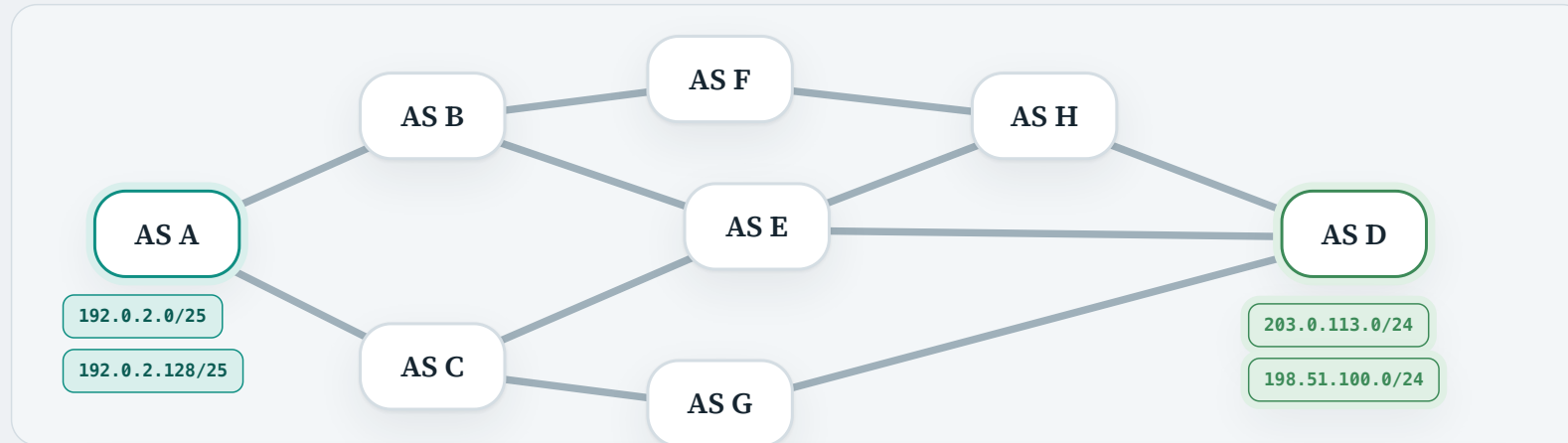
A quick recap of routing protocols

We can categorise a routing protocol by the **algorithm** underneath it — in Module 1 we met **distance vector** and **link state**.

We can also categorise it by **where it runs** — inside one autonomous system, or between them.

algorithm ↓ domain →	Intra-AS within one autonomous system	Inter-AS between autonomous systems
distance vector	RIP Routing Information Protocol next	
link state	OSPF Open Shortest Path First next	
path vector a close relative of distance vector		BGP Border Gateway Protocol now

The Internet is a network of autonomous systems



INTER-AS ROUTING PROBLEM

Now 192.0.2.1 sends instead to 203.0.113.42 — an address in **AS D**.

BGP is the Internet's inter-AS routing protocol

BGP is the Internet's answer to that problem. Here is how it does it.

LEARN EXTERNAL REACHABILITY

- Route-finding over a graph of **nodes** and **edges**, just as in Module 1 — except a node is no longer a router but a whole **autonomous system**, holding one or more subnets.
- Unlike Module 1, a destination is not a single IP address. It is a **destination prefix** in CIDR notation, like `203.0.113.0/24`, identifying a subnet inside some AS.
- For every **source–destination pair**, find a path across that graph — a path made of several ASes.

e.g. a path from AS A to AS D: **AS A → AS B → AS E → AS D**

DISTRIBUTE IT INSIDE THE AS

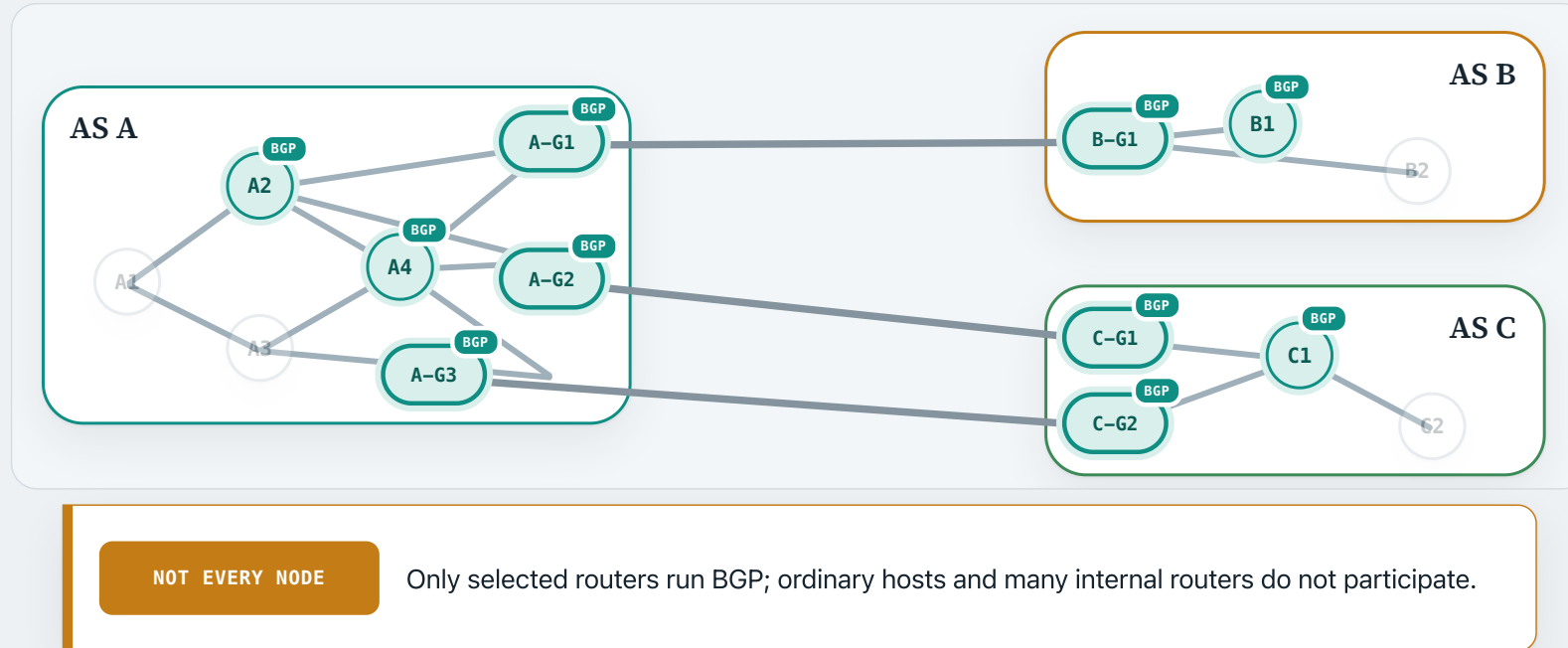
Once those external paths are known, the AS's own routers must be told, so they know how to forward packets leaving the AS.

SELECT A PREFERRED ROUTE

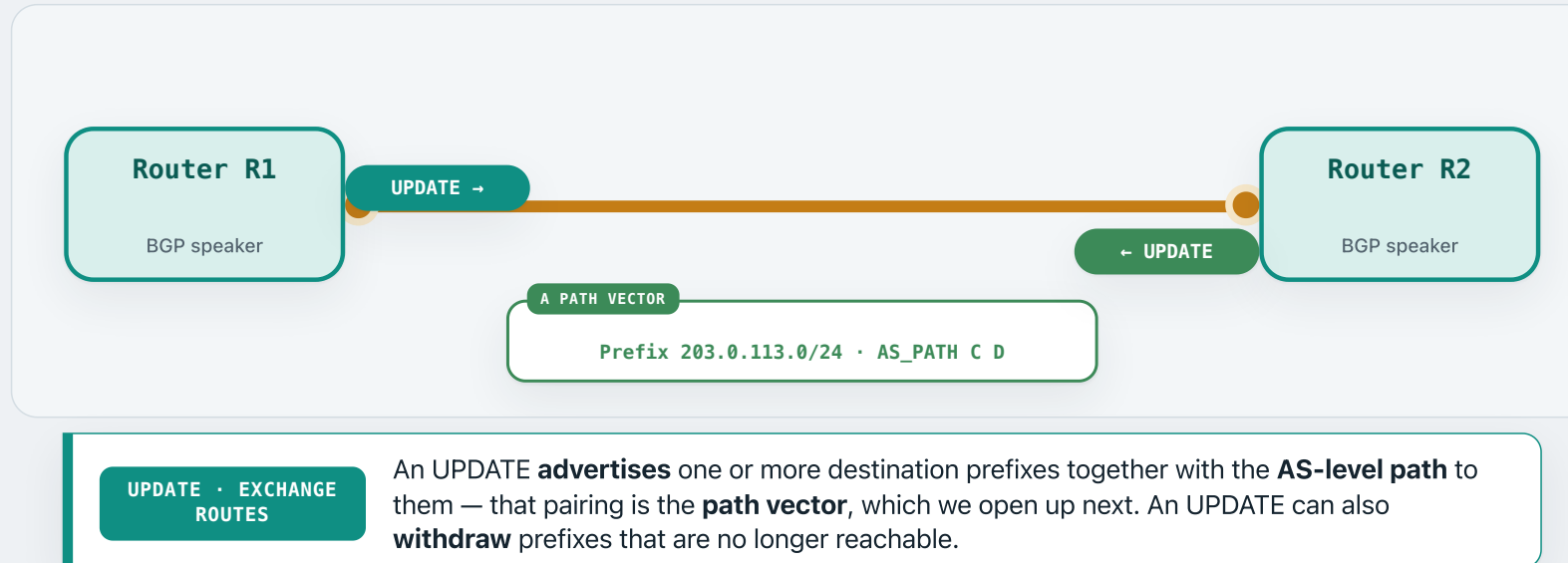
Again several paths may reach the same destination. In Module 1 the criterion was simple — the **shortest** path, one least cost. Here it is **multidimensional**: economic incentives, delay, and more. BGP supplies the criteria that pick the best one.

WHO PARTICIPATES IN BGP?

BGP is run by BGP-speaking routers

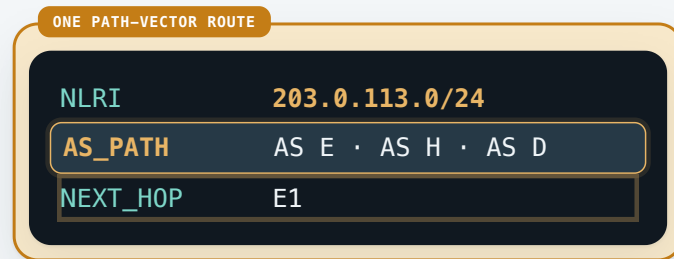


A BGP session is a structured conversation



A closer look at a BGP path vector

Unlike a distance-vector protocol, what BGP advertises to a neighbour is a path vector.



NLRI

Network Layer Reachability Information: the destination prefix being advertised.

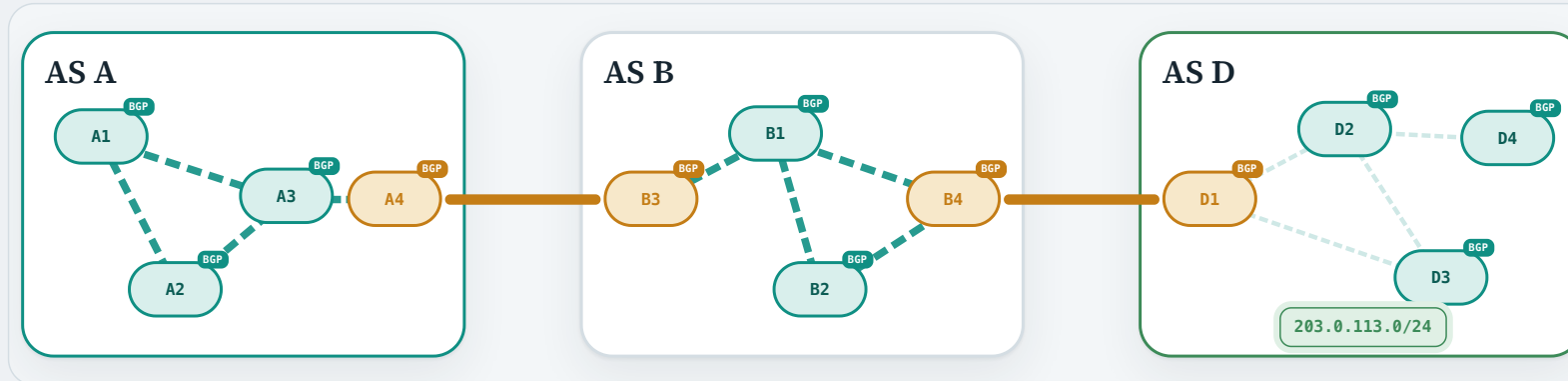
AS_PATH

The complete **AS-level path leading to that destination prefix**.

NEXT_HOP

The IP address of the next router to which traffic should be forwarded towards that prefix.

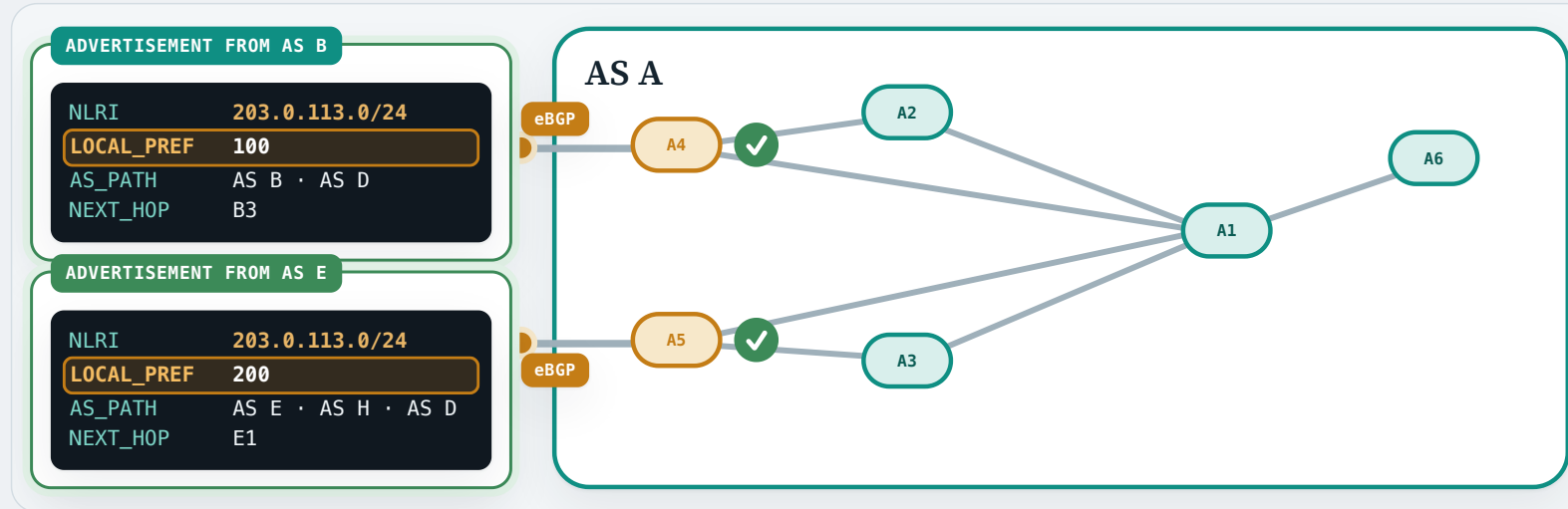
Watch one BGP route travel across three ASes



10 · EVERY ROUTER IN
AS A KNOWS THE WAY

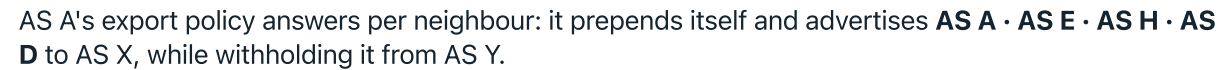
Every BGP speaker in AS A now knows that 203.0.113.0/24 is reachable **via AS B, then AS D** — carried between the ASes by **eBGP** and spread inside each one by **iBGP**.

eBGP import policy: which routes an AS prefers



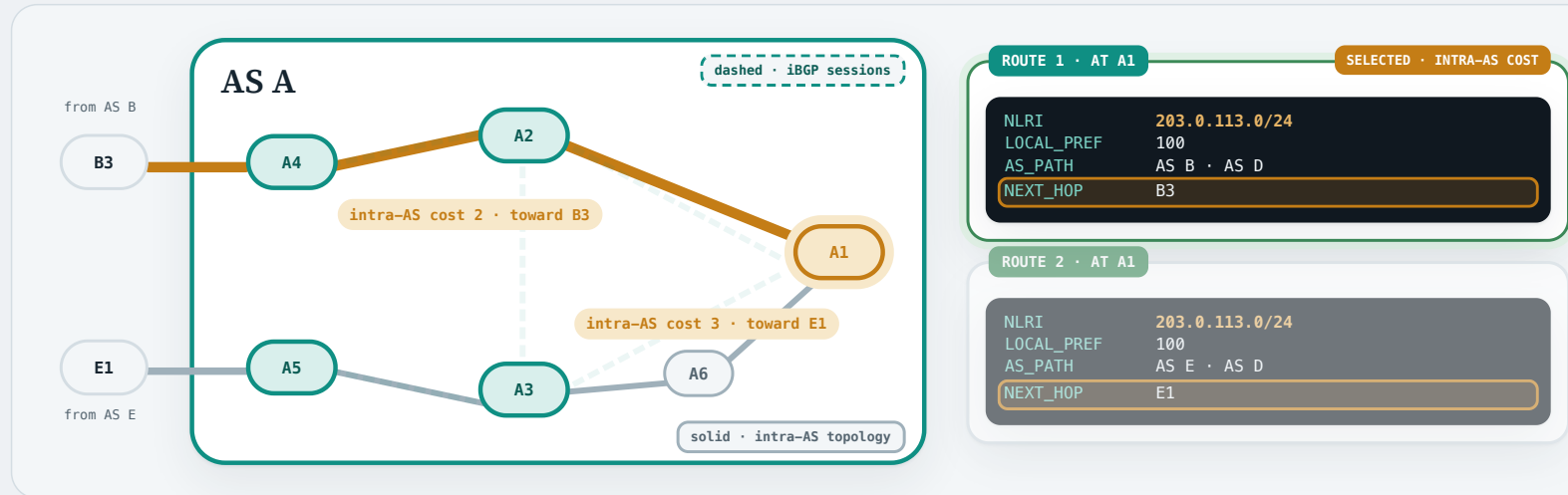
2 · ACCEPT BOTH, AND RANK THEM

AS A's import policy accepts both routes and attaches a LOCAL_PREF to each — 100 and 200 — for use inside AS A. The higher value is the one AS A prefers.



ONE NEXT HOP PER PREFIX · LOCAL_PREF → AS_PATH → INTRA-AS COST

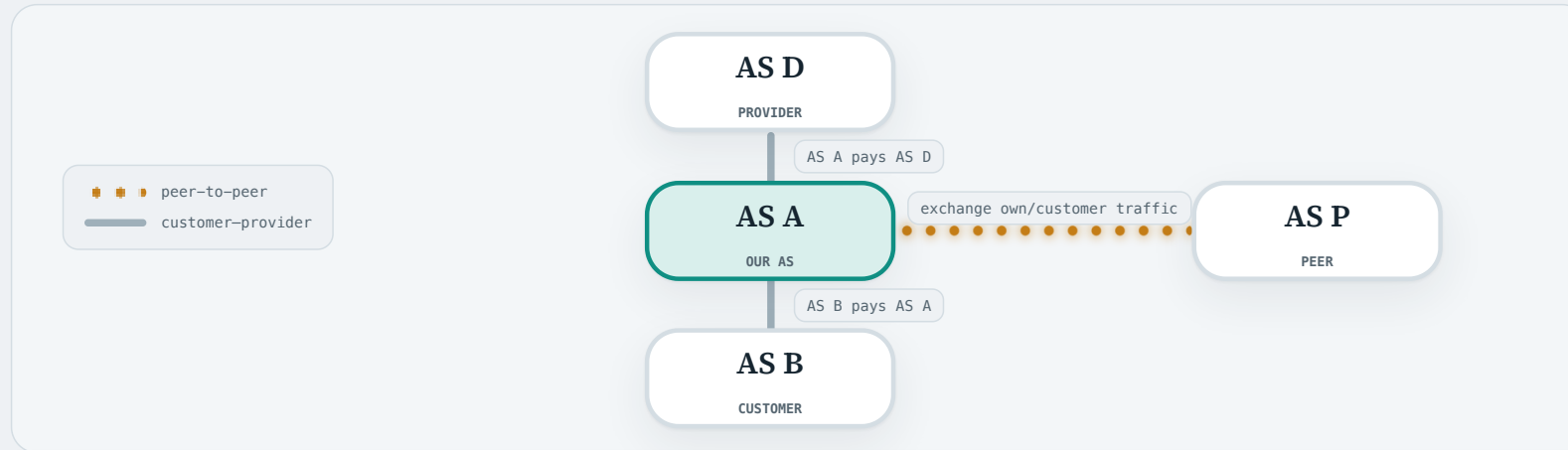
How each BGP router builds its forwarding table



3 · THEN HOT-POTATO ROUTING

If both BGP attributes tie, A1 turns to AS A's **own internal topology** and asks its own intra-AS routing protocol which next hop is closer: B3 at cost 2 beats E1 at cost 3, so **Route 1 wins — hot-potato routing**.

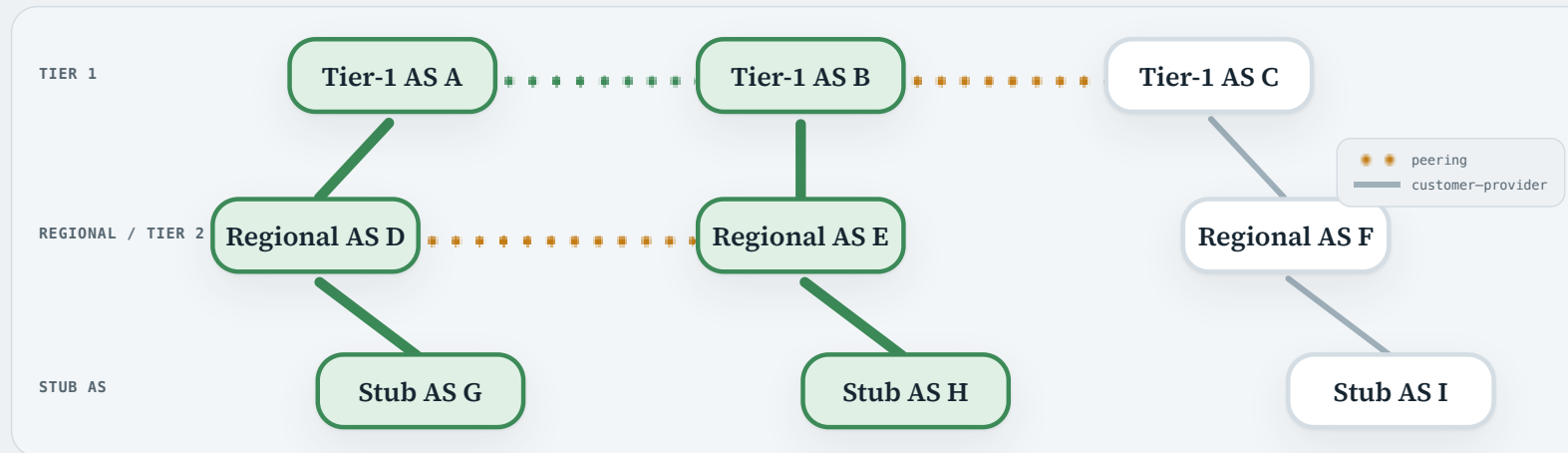
Business relationships between ASes



PEER ↔ PEER

Peers commonly exchange traffic for themselves and their customers—not general transit for one another.
e.g. AS A and AS P exchange roughly as much traffic as each other, so neither pays the other and neither depends on the other for connectivity.

These relationships create a commercial AS hierarchy



VALLEY-FREE PATHS

Whenever every AS's import and export policy is guided by these relationships, a path takes an **up-across-down** shape — a **valley-free path**.
e.g. a datagram from stub AS G to stub AS H climbs through providers to the Tier-1 level, crosses one peer link, then descends to AS H.

Import policy: which route should the AS prefer?

ROUTE LEARNED FROM	COMMON PREFERENCE	ECONOMIC INTUITION
Customer	HIGH	The customer pays us for transit.
Peer	MEDIUM	Usually settlement-free exchange.
Provider	LOW	We pay the provider for transit.

3 · PROVIDER ROUTE

Provider-learned routes commonly receive low preference because the AS must pay the provider to carry that traffic.

A COMMON COMMERCIAL GUIDELINE—NOT A RULE ENFORCED BY BGP

Export policy: where should the route be advertised?

ROUTE LEARNED FROM	TO CUSTOMER	TO PEER	TO PROVIDER
Customer	✓	✓	✓
Peer	✓	✗	✗
Provider	✓	✗	✗

PEER/PROVIDER ROUTES

Do not normally export these to another peer or provider; doing so creates unpaid transit.

Inter-AS routing recap

- **BGP solves the inter-AS routing problem** by learning AS-level paths to destination network prefixes.
- **eBGP exchanges routes between ASes**, while **iBGP distributes candidate routes inside an AS**.
- **Both use BGP's path-vector mechanism**, carrying a destination prefix together with its AS-level path and next hop.
- **Import and export policies express AS-wide routing choices**, often driven by customer, provider, and peer relationships.

MODULE 2 · ROUTING INSIDE ONE ADMINISTRATIVE DOMAIN

Intra-AS Routing Protocols

1 · Inter-AS routing and AS policy

2 · **Intra-AS routing protocols**

3 · Aggregation, NAT and IPv6

ROUTING WITHIN ONE AUTONOMOUS SYSTEM

An overview of intra-AS routing protocols

RIP

Routing Information Protocol

DISTANCE VECTOR

OSPF

Open Shortest Path First

LINK STATE

IS-IS

Intermediate System to
Intermediate System

LINK STATE

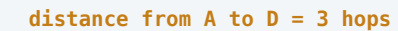
EIGRP

Enhanced Interior Gateway
Routing Protocol

DISTANCE VECTOR

Routers talk only to their direct neighbours, and from that alone each learns the **best distance** and the **next hop** to every destination. No router ever learns the full topology.

RIP measures distance in **router hops**: each router adds one to the metric it received.



A path of more than **15 hops** is treated as unreachable — a metric of 16 means infinity.

A RIP router keeps two tables

RIP routing table at A

destination	metric	next hop	timer
198.51.100.0/24	1	B	180 s
203.0.113.0/24	1	E	180 s
192.0.2.0/24	2	B	180 s



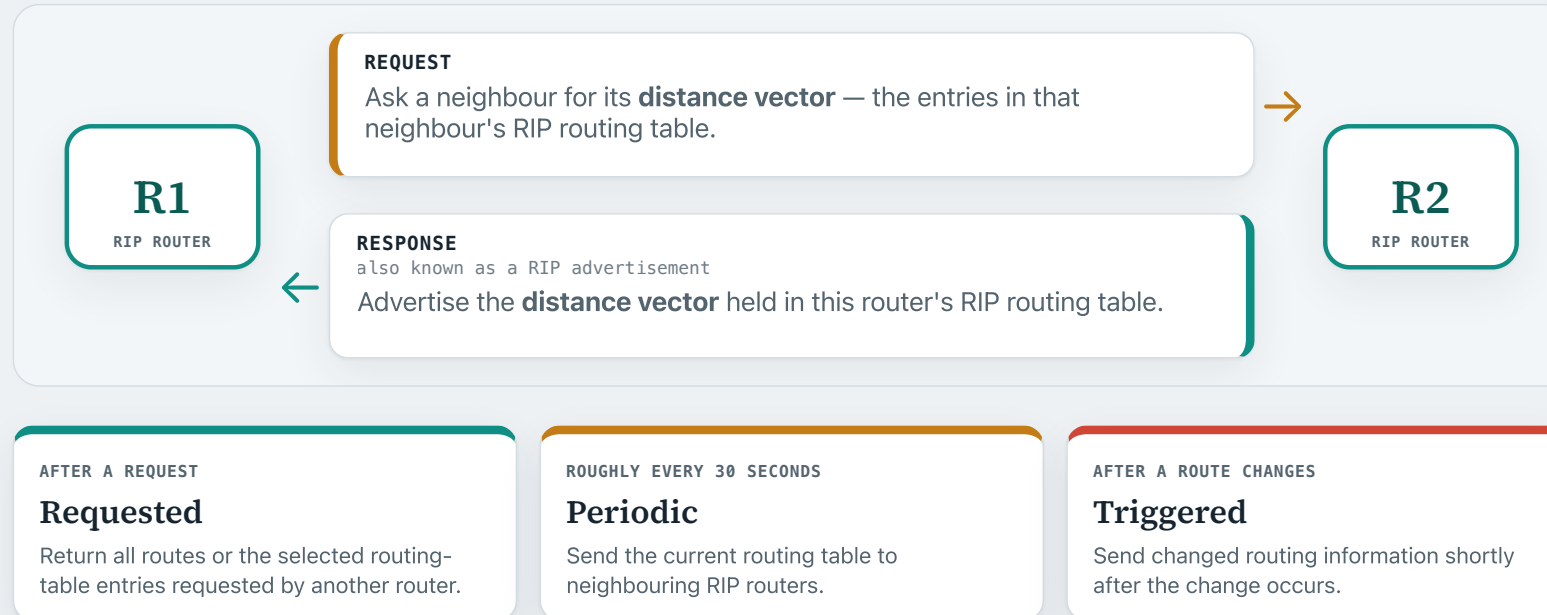
Forwarding table at A

destination	next hop
198.51.100.0/24	B
203.0.113.0/24	E
192.0.2.0/24	B

TWO TABLES, NOT ONE

They are kept separately, and may not always agree.
e.g. a route may be removed from the forwarding table, yet still be kept temporarily in the routing table.

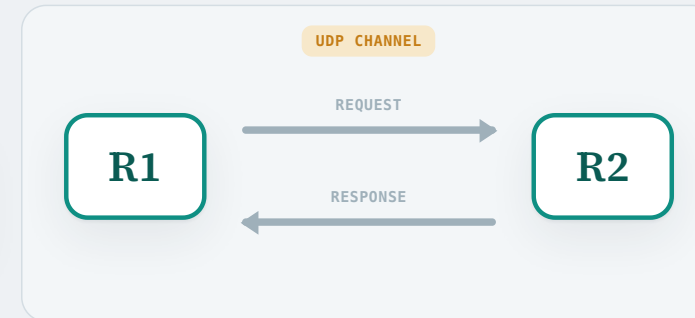
RIP carries Request and Response messages over UDP



An example of running RIP

RIP routing table at R1

destination	metric	next hop	timer
203.0.113.0/24	16	R2	120 s



7 · What if R2 stops sending them?

- **The timer runs out.** No valid refresh arrives from R2, and after 180 seconds the entry expires.
- **Mark it unreachable.** R1 sets the cost metric to **16** — RIP's infinity — and immediately drops the route from its **forwarding table**.
- **Keep it, to tell the neighbours.** The entry stays in the RIP routing table for another **120 seconds**, advertised with metric 16 so the neighbours learn the destination is gone.
- **Then remove it.** After those 120 seconds, R1 deletes the entry from its routing table altogether.

How RIP limits count to infinity

Recall from Module 1: a distance-vector protocol suffers from the **count-to-infinity problem**. RIP guards against it in three ways.

SPLIT HORIZON

Do not send it back

Never advertise a route back to the neighbour you learned it from. If R1's route to a destination came from R2, R1 leaves that destination out of the updates it sends to R2.

POISON REVERSE

Send it back as infinity

Stronger: R1 does advertise the route back to R2, but with a cost of **infinity**.

Both are only **heuristics**. They break a loop between **two** routers, but not one running through three or more.

COUNT ONLY TO 16

Make infinity deliberately small

Therefore, this is why in RIP the maximum path length is bounded to **15 hops**.

OSPF · Open Shortest Path First

What it is. OSPF is an intra-AS routing protocol which uses the **link-state** routing algorithm.

Protocol channel. OSPF carries its protocol packets directly over IP, without TCP or UDP.

A quick recap of the link-state routing protocol

- 1 Each router creates an **advertisement** for its local information, and tells the others.
↓
- 2 Every router uses all the advertisements to build the **same topology**.
↓
- 3 Each router runs a **shortest-path algorithm** over that topology to derive its forwarding table.

WHY CHOOSE OSPF?

OSPF provides useful features for a large AS

Authentication

OSPF can authenticate protocol exchanges so routers accept updates only from configured participants.

Operator-defined costs

The operator sets each link's cost, and so decides which paths OSPF prefers.

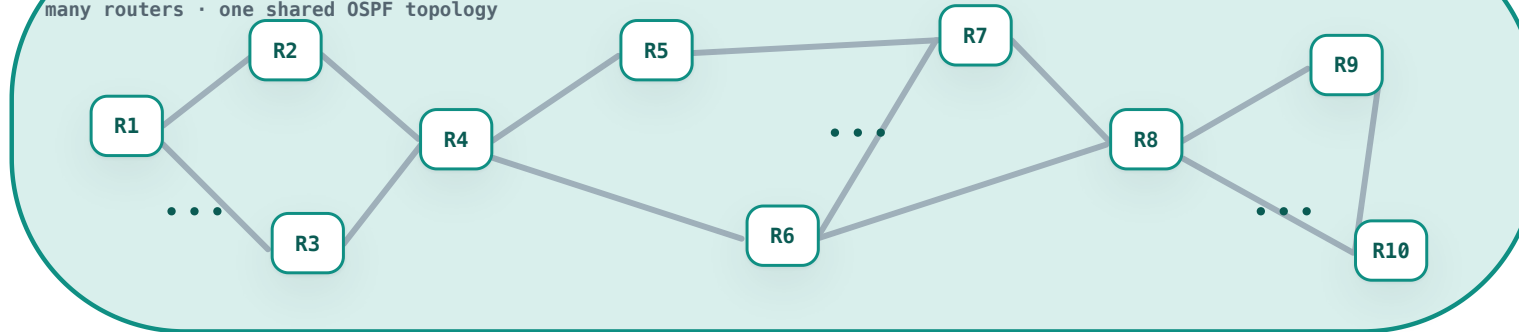
Area routing

A large AS can be divided into **areas**, so that OSPF scales.

Why can a large AS strain OSPF?

One large AS

many routers · one shared OSPF topology



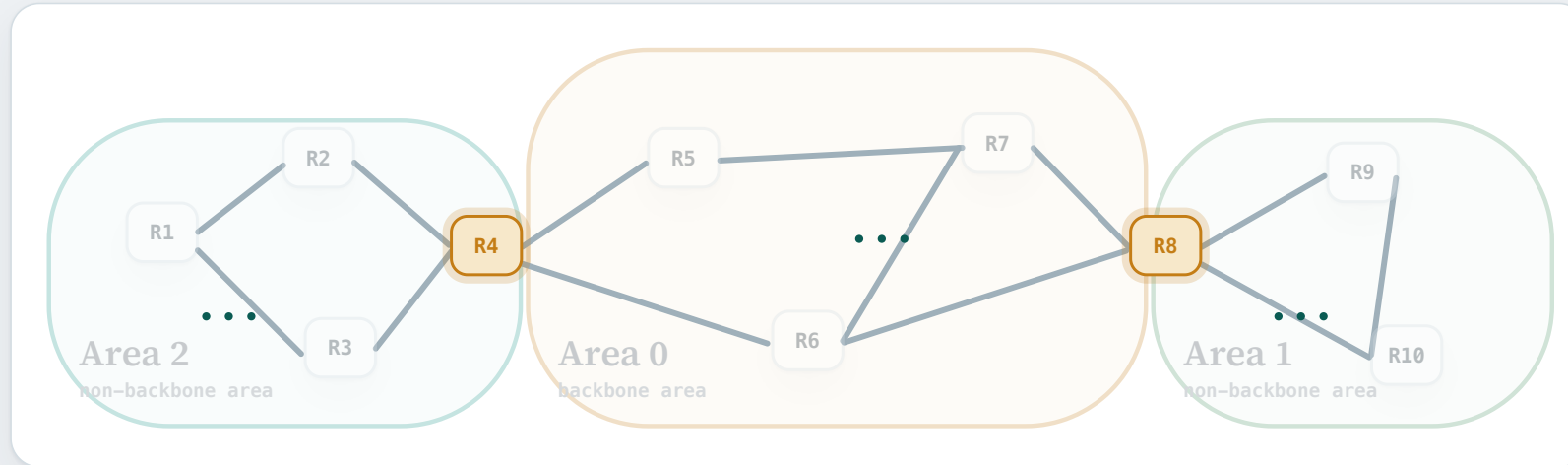
More memory

Every router stores the complete AS topology in its link-state database.

Slower convergence

Every change must be flooded widely before all routers calculate the new topology.

OSPF divides one AS into smaller areas



Non-backbone areas

Areas 1 and 2 each contain a smaller group of routers and networks.

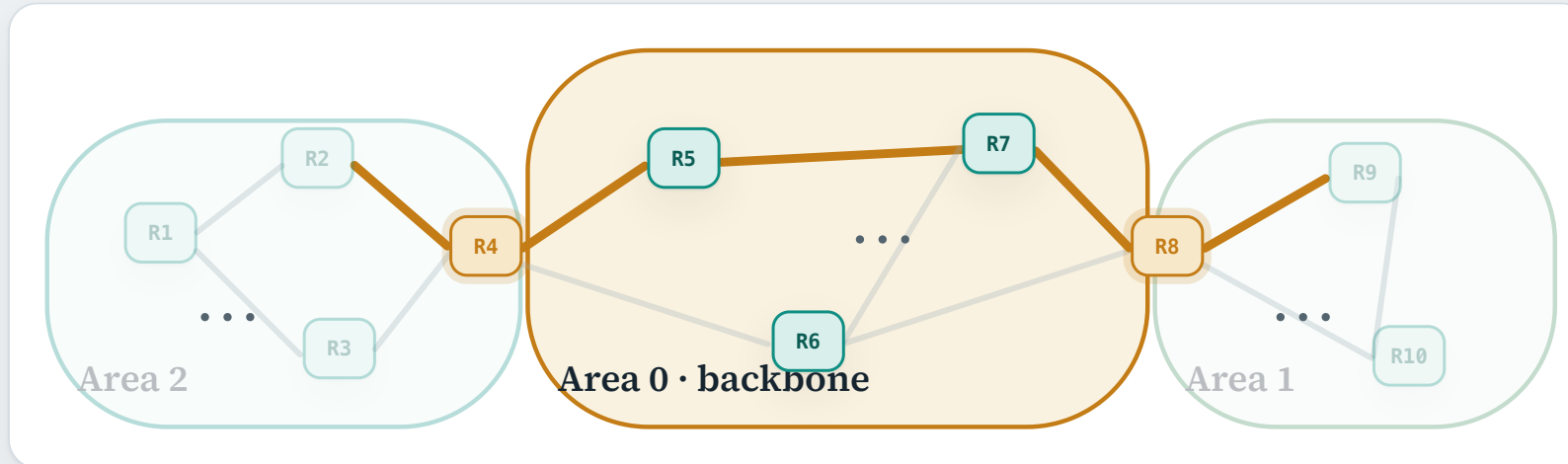
Backbone area

Area 0 connects the non-backbone areas and carries inter-area reachability.

Area border routers

R4 and R8 each participate in Area 0 and one non-backbone area.

How does hierarchical OSPF fit together?



Inside an area

- A router does its link-state flooding only within its own area, and stores that area's topology.
- Each router then runs the shortest-path algorithm over that topology to reach destinations inside its area.

Between areas

- ABRs carry condensed inter-area reachability through the backbone, Area 0.
- For example, a route from Area 2 to Area 1 can follow:
Area 2 → R4 → Area 0 → R8 → Area 1

Intra-AS routing recap



RIP is distance vector.

- It measures distance in router hops, and caps a path at 15.
- Neighbours exchange distance vectors, kept alive by periodic Responses.



OSPF is link state.

- Routers flood advertisements, so each builds the same topology.
- A large AS is divided into areas so that OSPF scales.

MODULE 2 · KEEPING THE INTERNET ADDRESSABLE

IP Addressing Scaling: Aggregation, NAT and IPv6

1 · Inter-AS routing and AS policy

2 · Intra-AS routing protocols

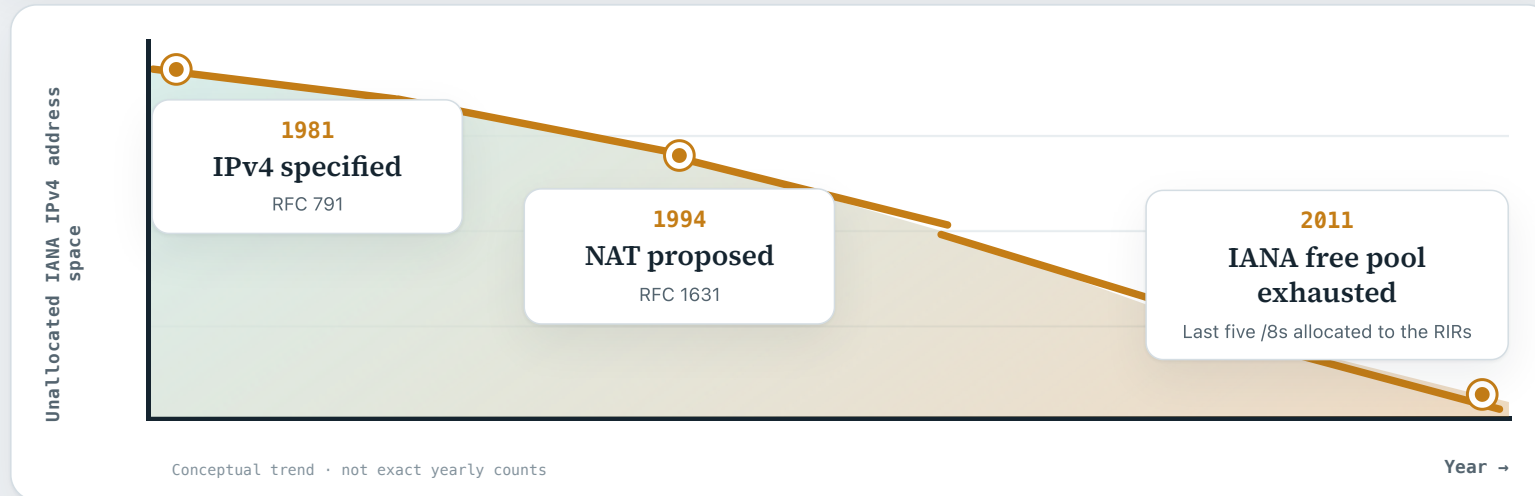
3 · Aggregation, NAT and IPv6

A BRIEF HISTORY

IPv4's address space became scarce as the Internet grew

2^{32} addresses

IPv4 uses 32-bit addresses. Its finite address space became an important limitation as more networks and devices joined the Internet.



How did the Internet adapt to IPv4 scarcity?

We will study three responses that extended IPv4's useful life and kept the Internet growing.

1

Classless Inter-Domain Routing

already covered in Term 3

2

Share public IPv4 addresses

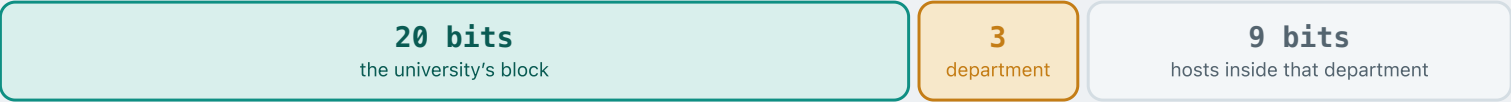
Network Address Translation (NAT)

3

Use a 128-bit address space

IPv6

CIDR stopped IPv4 address space going to waste



department	those 3 bits	its block
Engineering	000	200.23.16.0/23
Science	001	200.23.18.0/23
Law	010	200.23.20.0/23
Arts	011	200.23.22.0/23
...	...	...
Library	111	200.23.30.0/23

THE ANSWER

- Under **CIDR** the split can fall at **any bit**.
- e.g. a university takes **200.23.16.0/20** and divides it into eight sub-blocks: **3 bits** to say which department, and the remaining **9 bits** to say which host inside it.

IPv4 address aggregation

network	destination	next hop	in binary
the whole university	200.23.16.0/20	uc-gw	11001000 00010111 0001 0000 00000000

THE ANSWER

- The answer is **address aggregation**.
- In this example, all eight entries agree on their **first 20 bits** and share a next hop. So the eight can be replaced by one entry, **200.23.16.0/20**.

WHEN ONE NETWORK IS REACHED ANOTHER WAY

What if two entries both match?

network	destination	next hop	match for 200.23.18.5
the whole university	200.23.16.0/20	uc-gw	matches – 20 bits
Science	200.23.18.0/23	cloud-gw	matches – 23 bits

THE FORWARDING DECISION

- The Science department is now reached through a different provider, so its **next hop is different**. A second entry, **200.23.18.0/23**, is now inserted into the router's forwarding table.
- Now suppose a datagram arrives for **200.23.18.5**. It matches **both** destination prefixes — 20 bits on one, 23 on the other.
- **Which entry should the router use?**

LONGEST-PREFIX MATCH

When more than one entry matches a destination address, forward using the entry with the **longest** prefix.

Here 23 beats 20, so the datagram goes to **cloud-gw**.

MODULE 2 • IP ADDRESSING SCALING

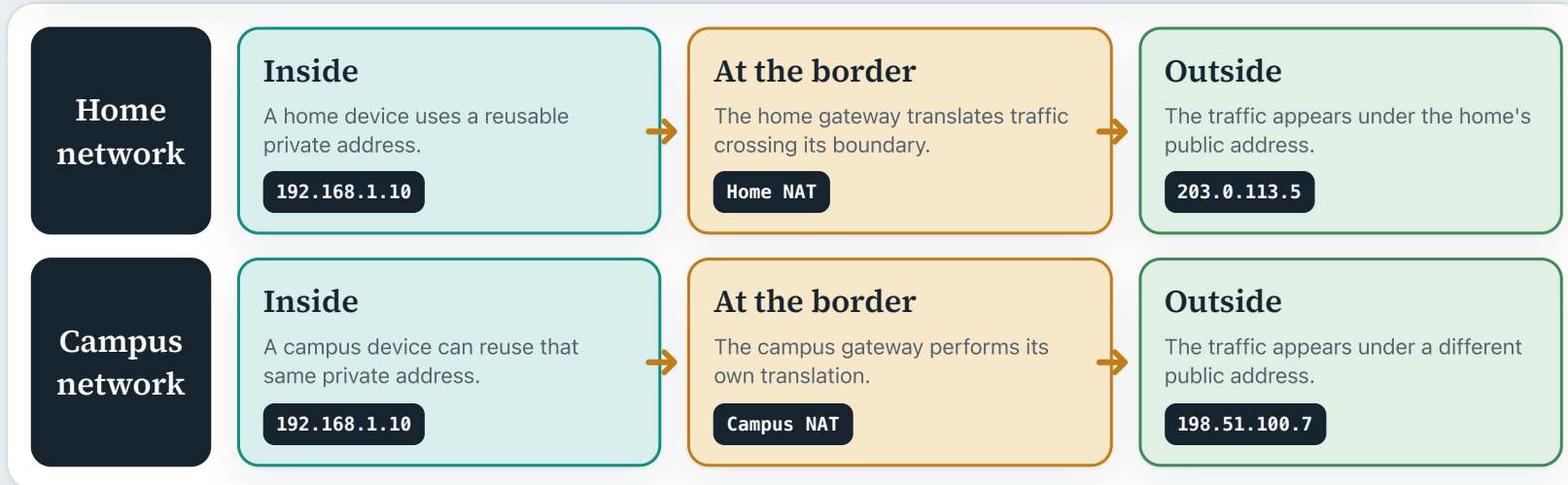
Network Address Translation

Why do so many local networks use addresses like 192.168.1.10?

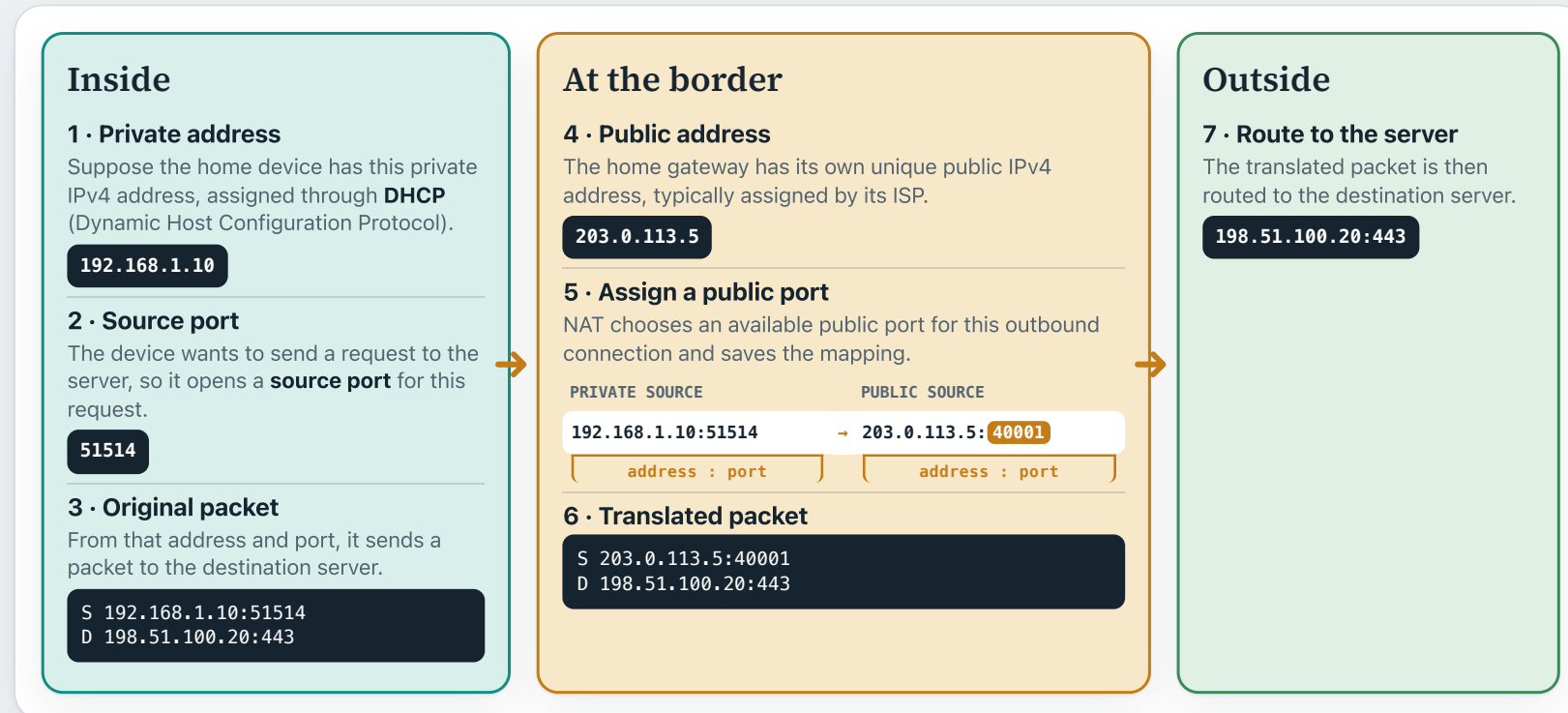
- **A familiar pattern.** You may see addresses such as 192.168.1.10 on devices at home, in an office, or in a campus lab.
- **Why is there no conflict?** If both your home and a campus lab contain 192.168.1.10, why does traffic sent at home reach the home device rather than the lab device?

NAT lets many private hosts share scarce public IPv4 addresses

As a result, the outside world only ever observes those two public addresses — and there is **no conflict**.



NAT translates both the address and the port



How does the reply reach the correct home device?



Advantages and disadvantages of NAT

Advantages

Address conservation: many devices can share one or a few public IPv4 addresses.

Addressing independence: internal addresses can change without affecting the public Internet, and the public address can change without renumbering every internal host.

Disadvantages

Per-flow state: the gateway must remember active translations.

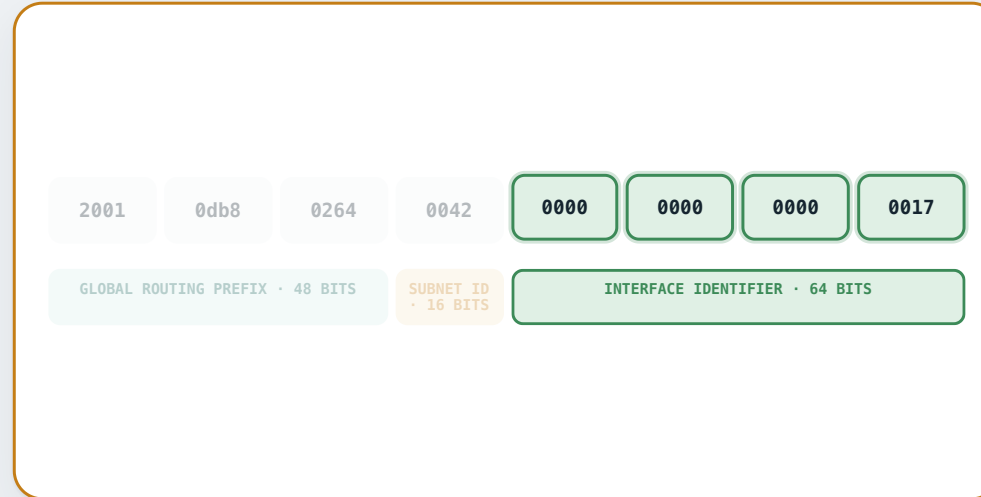
Inbound reachability: internal devices are not directly addressable from the public Internet. A mapping only comes into existence once the device has sent something out, so a reply can find its way back — nothing outside can start the conversation.

IPv6

- IPv6 is the **newer version** of the Internet Protocol.
- It uses **128-bit addresses**, far more than IPv4's 32-bit format.

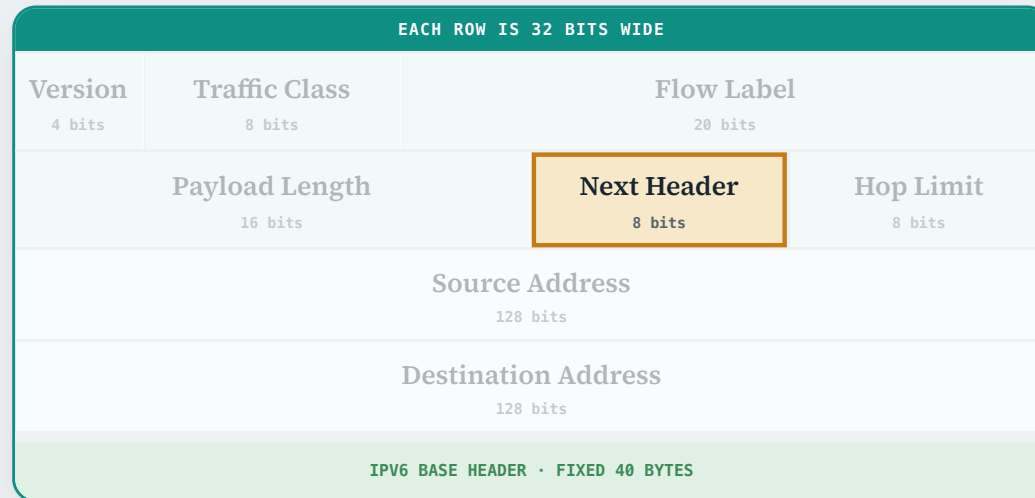
IPv6 address

- **First 48 bits • global routing prefix:** identify an allocated site or organization block. For example, a university could be allocated this block.
- **Next 16 bits • subnet ID:** identify one subnet inside that site. For example, this could be the Engineering department's network.
- **Final 64 bits • interface identifier:** identify one interface within that subnet. For example, this could be a laptop's Wi-Fi interface.



A breakdown of the IPv6 header

Every IPv6 packet begins with the same **fixed 40-byte base header**.



Source and destination addresses

Both fields are 128 bits and identify the packet's endpoints.

Hop Limit

Each router reduces it by one so a packet caught in a forwarding loop is eventually discarded.

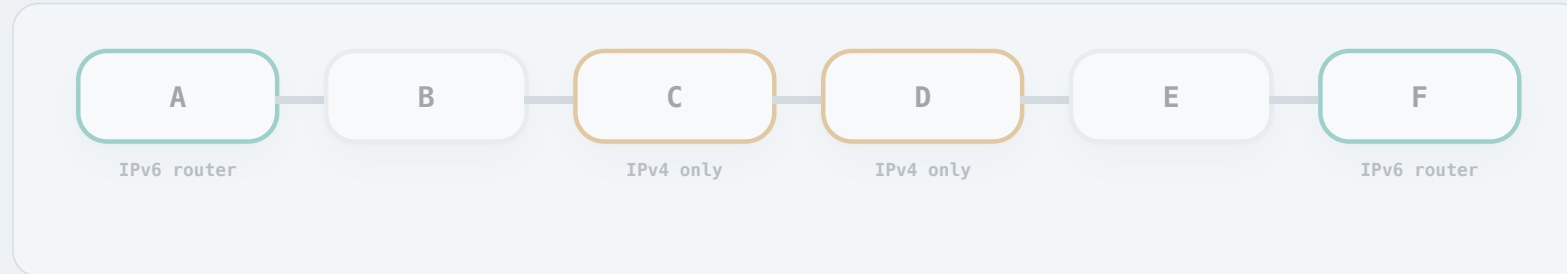
Flow Label

Marks packets belonging to the same flow — a TCP connection, for example — so routers can treat them alike.

Next Header

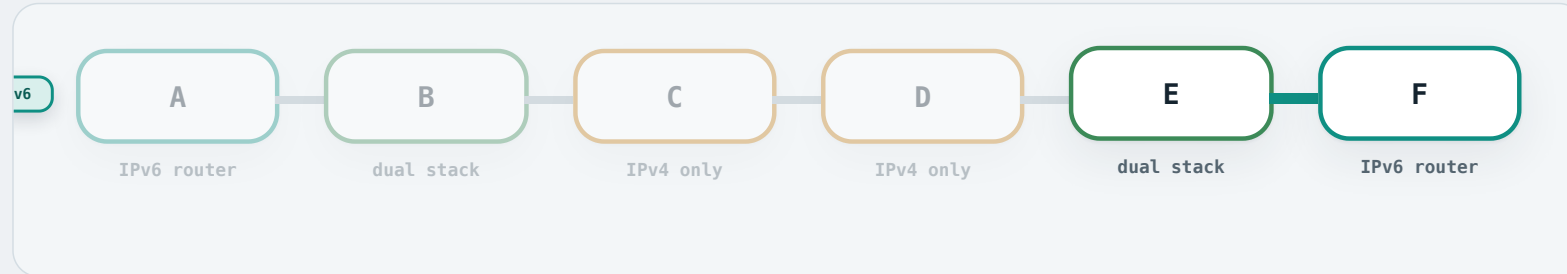
Identifies the extension header or upper-layer protocol that follows, such as a TCP or UDP header.

IPv6 traffic may encounter IPv4-only routers



- IPv6 is still being deployed, and progress has been slow: many routers in service are older commercial devices that support only IPv4.
- Suppose IPv6 router A wants to send an IPv6 packet to IPv6 router F.
- The path crosses IPv4-only routers C and D, which cannot forward that packet directly.
- **How can we resolve this compatibility problem?**

Two ways across the IPv4-only region



Delivery:

The unchanged IPv6 packet continues to F. Nothing was lost — encapsulation is **lossless**.



Module 2 recap

- **Inter-AS routing:** BGP learns AS-level paths, using eBGP between ASes and iBGP within an AS.
- **Intra-AS routing:** RIP uses distance vector, while OSPF uses link state and areas to route within an AS.
- **IP addressing:** NAT conserves public IPv4 addresses, and IPv6 expands the address space.

References

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Data and Computer Communications, 10th edition, Pearson, 2013 · Chapter 19, Routing.

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Computer Networks, 5th edition, Prentice Hall, 2010 · Chapter 5, The Network Layer.

James F. Kurose & Keith W. Ross

Computer Networking: A Top-Down Approach Featuring the Internet, 3rd edition, Addison Wesley, 2005 · Chapter 4, The Network Layer.

Core Internet standards

RFC 4271, 4632, 5737, 6598, 6888, 8200, and 4213, together with the protocol-specific RFCs cited on individual frames.

Number-resource authorities

IANA Number Resources and APNIC registry, allocation, and address-management material.

Teaching examples

Teaching topologies, addresses, and routing policies are explicitly constructed examples unless a frame identifies an external source.

Acknowledgements

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Shared CS340 lecture material used for several Internet routing examples.